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NON-RANDOM CRACK OPENING IN PARTIALLY CONFINED, THERMALLY DAMAGED PBX 9501 AND OBSERVATIONS ON ITS EFFECTS ON COMBUSTION

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Abstract. In this work, we present evidence for how strong radial confinement can result in aligned macro-scale crack opening. We damage cylinders in a tight-fitting quartz sleeve, open on both ends, and observe the occurrence of aligned cracks opening normal to the longitudinal axis. This geometry and confinement is common in experimental arrangements such as strand burners and DDT tubes. Further, we observe, with high-speed photography, how this non-random crack opening affects combustion, and propose mechanisms, garnered from time-lapse photography and elastic stress analysis, for how it occurs.

Keywords: Thermal damage, explosives, combustion, cracks, porosity

PACS: 82.33.Vx, 82.33.Ln

INTRODUCTION

PBX 9501 is a plastic-bonded explosive composed of (by mass) HMX crystals (95%) coated in a polymeric binder (2.5% estane and 2.5% BDNPA/F plasticizer) and pressed to within 98.5% of theoretical maximum density, which undergoes complex changes when heated to high temperatures. For example, above 77°C the binder melts and near 120°C, the plasticizer begins to gasify [1]. Above 160°C, the HMX crystals begin to go through a polymorphic β - δ phase transition [2] with an accompanying 7% increase in volume and significant intragranular fracturing [3]. At temperatures above 180°C, appreciable HMX decomposition and gasification begins to occur [4]. Together, all of these thermally induced changes lead to complex morphological alterations that can be broadly described as “damage”.

Damage, particularly the creation of cracks, has been shown to modify the combustion mode and violence of burning explosives, since flames can intrude at lower pressures through cracks than through the more tortuous matrix porosity [5, 6].

However, for fluid transport processes, including convective combustion and gas permeation, to occur the porosity must be *open* and accessible to the surfaces of the charge. Because many charges are encased (confined), there are mechanical limitations on the extent to which thermally induced cracks can open. Understanding how confinement affects crack formation and its subsequent influence on transport may prove to be important.

In this study, we have thermally damaged PBX-9501 at 185°C for durations of 1-3 hours and observed both micro- and macroscopic cracks. We examined how changing the tightness of the sample confinement changes both the micro- and macroscopic damage patterns. We demonstrate that these damage patterns affect flame intrusion characteristics.

EXPERIMENTAL

A thick-walled quartz tube (ID of 8.20 mm, OD of 38 mm $\times$ 30 mm) was machined to accept the

high explosive (HE) samples. Next, a copper sleeve with a slot to allow optical access along the length was fitted around the OD of the quartz tube. The copper sleeve had resistive heater rods inserted with thermocouples to monitor and control temperature. The complete assembly allowed samples to be thermally damaged and observed *in situ* in a variety of experiments (Fig. 1).

Cylinders of PBX-9501 (7.88 or 8.18 mm $\varnothing \times$ 20-30 mm) were inserted into the quartz tube so that they were unconfined longitudinally and confined radially around their circumferences. The smaller diameter cylinders fit “loosely” and were free to expand 3.9% diametrically before coming into firm contact with the confining quartz tube. The larger diameter cylinders fit “tightly” and were free to expand only 0.2% diametrically.

Time-lapse photography and closed-vessel combustion studies were performed on the damaged samples. For the time-lapse photography, the samples were heated at ambient pressure. For the closed-vessel combustion, the entire assembly was placed and sealed inside a 2-liter combustion bomb that would be pressurized to varying extents (between 8 and 15 MPa) 1 minute before sample ignition by hot wire (more details can be found in Ref. 6). Pressure was monitored inside the closed-bomb. High-speed photography was performed for some of the closed-vessel burns. Finally, reflected polarized light microscopy (PLM) was performed on PBX-9501 cylinders damaged in the same way (this technique is described in detail in Ref. 7).

RESULTS AND DISCUSSION

Observations from time-lapse photography

The first sample was a tight-fitting cylinder *with* vacuum grease lubricating the HE/quartz interface. Image sequences taken over the span of 1 hour reveal axial elongation followed by macroscopic, radially oriented cracks forming and then opening near the ends of the cylinder. One of the cracks contains amber-colored molten binder in which gas bubbles were seen repeatedly forming and rupturing. The cracking occurred after 55 minutes at 185°C (Fig 2). The second sample was a tight-fitting cylinder *without* lubrication. Similar to the sample with grease lubricating the HE/Quartz

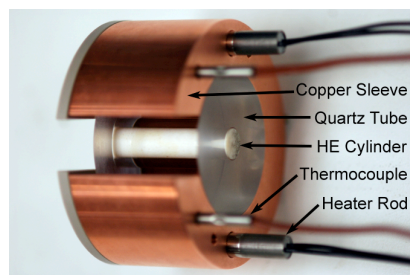


Figure 1. The assembly.

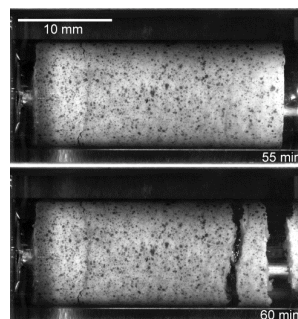


Figure 2. Two frames from time-lapse photography showing the opening of radially oriented cracks near the end of the cylinder. This sample was tight-fitting and lubricated.

interface, this sample cracked and separated with the same radial orientation and also near the end of the cylinder. However, the crack formed 90 seconds after the sample reached the target temperature of 185°C, but before the axial elongation was complete. After 90 minutes at elevated temperature two more cracks with the same orientation were revealed as amber-colored molten binder migrated into them. They do not however separate from the remainder of the cylinder as seen in the previous sample. The third sample was a loose-fitting cylinder with lubrication. Image sequences reveal the sample swelling in all directions, but without the formation of macroscopic cracks.

Closed vessel combustion

Tight-fitting samples with grease lubricating the HE/quartz interface were damaged at 185°C for 3 hours and then ignited inside a combustion bomb. High-speed video shows three characteristic burn modes. The first mode can be described as laminar

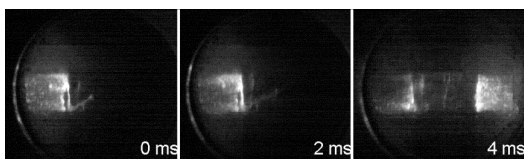


Figure 3. Frames from high-speed video of a closed bomb combustion experiment showing illumination of radially oriented cracks as flame penetrates the cylinder. The cylinder was ignited on the left side. This burn mode is convective.

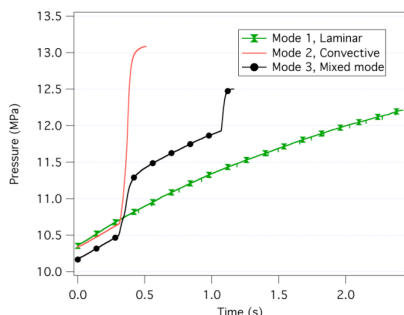


Figure 4. Representative pressure histories showing the three modes of combustion observed.

burning where the burn front progresses down the longitudinal axis of the sample in a relatively slow, steady layer-by-layer fashion. In this mode, the flames do not intrude or access the internal porosity. Pristine samples burn in this fashion.

The second characteristic burn mode is best described as convective (or deconsolidative) where the flames can be seen to illuminate radially-oriented cracks in a rapid, often unsteady, progression. With the convective burn mode, multiple aligned, cracks were sometimes observed to ignite simultaneously (within a 2 ms interframe time) along the longitudinal axis of the sample, followed by deconsolidation at the crack boundaries (Fig. 3). This characteristic mode is observed in thermally damaged samples and is in qualitative agreement with the aligned crack opening observed in the time-lapse sequences and the microscopy.

Finally, we have taken video of samples where the burn will switch from convective to laminar and then back to convective as it progresses down the longitudinal axis of the sample. This is the third mode, but in actuality is a combination of the previous two and is interpreted to indicate partially cracked samples where macroscopic cracks opened

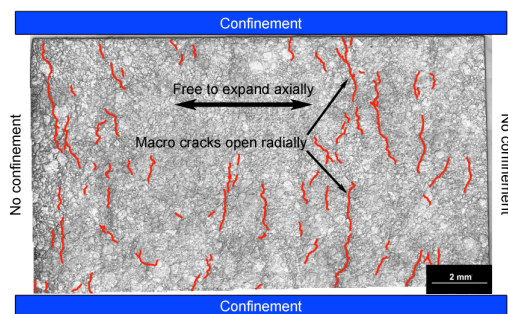


Figure 5. A mosaic of PLM micrographs with the widest cracks highlighted for emphasis.

only near the ends of the sample cylinders. Any cracking that may have occurred near the middle was not accessible to flame intrusion at these pressures.

The burn rate and accessibility of surface area of a burning sample can be estimated from the recorded pressure history, as combustion product gases are liberated inside a combustion bomb. Increasing rates of pressurization of the vessel indicate an increasing amount of surface area incorporated in the burn. Again, in agreement with the high-speed video, we can classify the pressure-time records into the same three modes (Fig. 4).

Microscopy

Microscopic examination by PLM of the thermally induced cracking in both loosely and tightly confined samples reveals damage patterns in agreement with the macroscopic observations from the time-lapse experiments. Figure 5 is a mosaic of multiple high-resolution micrographs where the widest (most open) cracks have been highlighted. This illustrates the aligned nature of the crack opening seen only in samples damaged in tight radial confinement. The microscopic analysis is in agreement with the other methods of observation discussed in this work.

Microscopic analysis shows that, regardless of the tightness of confinement, micro-scale cracks form during thermal insult. However, macroscopic analyses show that multiple, aligned macro-scale cracks open with a specific orientation in samples where the confinement is tight but with room for expansion in one dimension. This type of radial confinement is commonly employed in many types



Figure 6. A piece of thermally damaged PBX 9501 (from a different experiment) exhibiting similar crack patterns. The confinement for this experiment was nominally similar. Note: the units on the ruler are mm and its length is 90 mm.

of experiments and the resulting damage patterns appear to be independent of scale (Fig. 6).

Elastic stress analysis showed that the stress fields during heating in radial confinement would only be compressive. We speculate that there are two different mechanisms that cause cracks to open in this orientation. First, in the tests with grease lubricating the HE/quartz interface, localized pressure due to plasticizer gasification may cause the cracks to open. Conversely, with no grease, the frictional interaction with the quartz walls may lead to a stick-slip mechanism and thus transient tensile stresses. In this case the early crack opening was roughly in conjunction with the rapid expansion due the phase change (shortly after the sample reached 180°C).

High-speed video relates unsteady convective burning (and overall faster combustion) to the observation of macroscopic cracks. Researchers [8] have shown that flames can penetrate macro-scale cracks at relatively low pressures and then spread at rates over 2 km s^{-1} . Further evidence shows gas choking [9] and subsequent “combustion bootstrapping” can lead to rapidly increasing pressurization and runaway incorporation of increasingly finer matrix porosity in combustion [10, 11]. Together, these accelerating processes support each other and may increase the likelihood of DDT.

CONCLUSIONS

We have learned that the understanding of crack formation and its influence on combustion is

a complex problem. Further understanding of the mechanical properties of hot PBX-9501 is needed. For now, we have shown that cracking exists on varying length scales and orientations dependent on a large number of variables and that tight, partial confinement will lead to macroscopic cracks and more violent combustion.

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